EchoKey Seven-Operator Structural Analysis of the Kim-Holonomy Open Problem

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Abstract

We give a seven-operator structural analysis of the open problem (OP2 of [9]) of lifting the abelian Collatz holonomy of [9] to a $\mathbb{Z}_p$-valued invariant in Kim's arithmetic Chern–Simons framework. Using the seven operators of the EchoKey v2 specification [7] — Cyclicity, Recursion, Fractality, Regression, Synergy, Refraction, Outliers — we assign each operator a specific object in the Kim/Collatz setting and test whether the operator's prediction reproduces a measured anchor from [9]. All seven assignments produce predictions matching their anchor within pre-declared tolerances; in particular, the Outliers operator independently recovers the Orzechowski crossing location at step $k^* = 76$ from the data, against the Paper 1 measurement of step 75, with no input from the location measurement. The seven-operator decomposition realises the three structural requirements of OP2 (basepoint, p-adic Lie group, groupoid morphisms) and is consistent with every Paper 1 empirical constraint. We frame this as structural consistency between the EchoKey and Kim frameworks on the Collatz test bed, not as an internal proof in Kim's framework, and outline what the upgrade to such a proof would require.

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1 Introduction

1.1 Background and Open Problem

In [9] the abelian Collatz holonomy $\mathrm{hol}_{\mathrm{add}}(\gamma_n) = \sum_{j=0}^{|\gamma_n|-1} v_2(\gamma_n^{(j)})$ was introduced as a candidate scalar shadow of an underlying $\mathbb{Z}_p$-valued holonomy in Kim's arithmetic Chern–Simons theory [5]. Empirical computation gave Pearson correlation $r = 0.96$ of $\mathrm{hol}_{\mathrm{add}}$ with trajectory length, normalized mean 1.22 over $n \leq 100$, an attractor signature $\mathrm{hol}_{\mathrm{norm}}(\{1, 2\}) = 1/3$, tail-holonomy convergence to 1.18 ± 0.18, a robust supermartingale signature across six scales of N, and the Orzechowski crossing as a sharply localized late-trajectory phase transition at support size 4 with $\Delta\bar{v}_2 = -0.48$.

Open problem OP2 of [9] asks for the lift: specify (i) the basepoint in $\pi_1^{\mathrm{un}}(\mathrm{Spec}(\mathbb{Z}_2))$, (ii) the target p-adic Lie group, and (iii) the sense in which discrete Collatz steps define morphisms in the étale fundamental groupoid, such that the resulting holonomy reduces to $\mathrm{hol}_{\mathrm{add}}$ in the abelian quotient and is consistent with the empirical constraints listed above. The present paper does not solve OP2 in Kim's framework. It instead decomposes the structure that OP2 demands using a separate operator framework — the EchoKey v2 specification of [7] — and tests whether the seven-operator decomposition is consistent with every Paper 1 measurement.

1.2 Approach and Verification Framing

The EchoKey v2 specification fixes seven operators acting on a separable Hilbert space $\mathcal{H}$:

$\mathcal{C}$ (Cyclicity), $\mathcal{R}$ (Recursion), $\mathcal{F}$ (Fractality), $\mathcal{G}$ (Regression), $\mathcal{S}$ (Synergy), Π (Refraction), $\mathcal{O}$ (Outliers),

each with a formal definition (Sections 3–9 of [7]). For each operator, we (a) assign it a specific object in the Kim/Collatz setting, (b) compute a numerical prediction using only the operator's formal definition, and (c) compare to a measured anchor from [9]. A test is a *match* when the prediction lies within a pre-declared tolerance of the measurement; otherwise it is a partial match or a mismatch.

This is a structural-consistency analysis, not a theorem in Kim's framework. A successful seven-out-of-seven match shows that EchoKey's operators, applied via the proposed assignments, do not contradict any Paper 1 measurement and recover the empirical structure of OP2's constraints. It does not show that EchoKey's operators *are* the correct decomposition of Kim's framework. The upgrade from structural consistency to a Kim-internal proof is identified as an open problem in Section 6.

1.3 Acknowledgments of Frameworks

This paper builds on the $\mathrm{M}^2\mathrm{KR}$ dictionary of arithmetic topology [1, 4, 2, 3], Kim's arithmetic Chern–Simons theory [5], the empirical Collatz transfer-operator framework of [9], and the EchoKey v2 question-first specification of [7]. The applied EchoKey-EFECGSC work of [8] provides the simulation context in which six of the seven operators were first deployed in concrete computation; the seventh operator (Refraction) is taken from the v2 specification.

2 The EchoKey v2 Operators

We collect the seven operator definitions used in this paper, retaining the notation of [7]. Throughout, $\mathcal{H}$ is a separable Hilbert space and $\Psi \in \mathcal{H}$ a state.

Definition 2.1 (Cyclicity [7], Def. 3.1)**.** *For period $T > 0$ and angular frequencies $\omega_n = 2\pi n/T$,*

$$\mathcal{C}_T[f](t) = \sum_{n\in\mathbb{Z}} \left(\frac{1}{T} \int_0^T f(\tau) e^{-i\omega_n \tau}\, d\tau \right) e^{i\omega_n t}.$$

$\mathcal{C}_T$ is the Fourier-like projector onto the period-T subspace of $L^2([0,T])$.

Definition 2.2 (Recursion [7], Def. 4.1)**.** *For a map $\mathcal{F} : \mathcal{H} \to \mathcal{H}$, the iterated functional transform is $\mathcal{R}^{(n)}[f] = \mathcal{F}(\mathcal{R}^{(n-1)}[f])$ with $\mathcal{R}^{(0)}[f] = f$, and $\mathcal{R}[f] = \lim_{n\to\infty} \mathcal{R}^{(n)}[f]$ when the limit exists. Existence is guaranteed by Banach contraction whenever $\mathcal{F}$ is a contraction on a metric subset of $\mathcal{H}$.*

Definition 2.3 (Fractality [7], Def. 5.1)**.** *For a dilation $D_\lambda \phi(x) = \phi(\lambda x)$ with $\lambda > 1$, weight sequence $\{\alpha_k\}_{k\geq 0}$, and a map $\mathcal{E}$ on $\mathcal{H}$,*

$$\mathcal{F}[\mathcal{E}](\Psi) = \sum_{k=0}^{\infty} \alpha_k\, \mathcal{E}(D_{\lambda^k}\Psi).$$

Definition 2.4 (Regression [7], Def. 6.1)**.** *For a reference state $\Psi_\star \in \mathcal{H}$ and rate $\lambda > 0$,*

$$\mathcal{G}_\lambda[\Psi](t) = \Psi_\star + e^{-\lambda t}\big(\Psi(0) - \Psi_\star\big).$$

Definition 2.5 (Synergy [7], Def. 7.1)**.** *For components $\Psi = (\Psi_1, \ldots, \Psi_n) \in \mathcal{H}^n$, coefficients κ_{ij}, and a bilinear form B,*

$$\mathcal{S}[\Psi] = \sum_{1\leq i<j\leq n} \kappa_{ij}\, B(\Psi_i, \Psi_j).$$

Definition 2.6 (Refraction [7], Def. 8.1)**.** *For a layer index $L \in \mathbb{N}$ and a transform density $\eta : \mathcal{H} \to \mathcal{H}$,*

$$\Pi[\Psi; L] = \Psi + \mu L\, \eta(\Psi),$$

interpreted as a small deformation across an interface or domain change.

Definition 2.7 (Outliers [7], Def. 9.1)**.** *For times and weights $\{(t_k, w_k)\}_{k\in\mathbb{N}}$ and jump directions $\Xi_k \in \mathcal{H}$,*

$$\mathcal{O}[\Psi](t) = \Psi(t) + \sum_k w_k\, \delta(t - t_k)\, \Xi_k.$$

3 Operator Assignments

For each operator we specify the Kim/Collatz object to which we assign it and the Paper 1 empirical anchor against which we test. The full table:

Op	Name	Assigned Collatz/Kim object	Paper 1 empirical anchor
$\mathcal{C}$	Cyclicity	Attractor cycle $\{1,2\}$ realized as $[1,2,1]$	$\mathrm{hol}_{\mathrm{norm}}(\{1,2\}) = 1/3$
$\mathcal{R}$	Recursion	Iterated transfer operator $(T_3)_*$ on μ_k	Tail-holonomy std contraction rate $\alpha < 1$ (Banach condition)
$\mathcal{F}$	Fractality	Multiscale $\mathbb{Z}_2 = \varprojlim \mathbb{Z}/2^n$, $\lambda = 2$, $\alpha_k = 2^{-k}$	Pearson $r(\mathrm{hol}_{\mathrm{add}}, \lvert\gamma\rvert) = 0.960$
$\mathcal{G}$	Regression	Cesàro mean $C_k = \frac{1}{k+1}\sum_{j\le k}\mathbb{E}_{\mu_j}[f]$	Steady-state $f_\star \in [0.4, 0.8]$ and decay rate $\lambda > 0$
$\mathcal{S}$	Synergy	Parity-branch bilinear coupling $B(\Psi_{\mathrm{odd}}, \Psi_{\mathrm{even}})$ at the Orzechowski crossing	$M_1 = \lvert\mathbb{E}[f\lvert B{=}1] - \mathbb{E}[f\lvert B{=}0]\rvert = 0.625$
Π	Refraction	Siegel (p,q)-adic conjugacy as $L = 1$ layer transform	v_2-mean invariance: max layerwise difference $= 0$
$\mathcal{O}$	Outliers	Orzechowski crossing as singular impulse in v_2-mean	Impulse location k^* and amplitude $\Delta\bar{v}_2 = -0.48$

The assignments are not forced — alternative assignments would be possible — but each is principled in the sense that the Collatz/Kim object selected is the natural target of the operator's formal definition. The assignment is recorded here in advance of the test; the verification proper is in Section 4.

4 Verification Results

All results in this section were produced by `collatz_echokey_probe.py` (CC0, available at [9]), which depends on `collatz_arithmetic_probe.py` for its Paper 1 anchor data. We report each test in turn.

4.1 Cyclicity: Attractor Holonomy

The attractor cycle $1 \to 2 \to 1$ realized as $[1,2,1]$ has v_2-sequence $[0,1,0]$. The discrete Fourier projector onto period-3 sequences is the identity on length-3 vectors; idempotence is satisfied to floating-point precision ($\|\mathcal{C}_T^2 x - \mathcal{C}_T x\| = 7.4\times 10^{-17}$). The substantive prediction is the time average of v_2 over one period:

$$\frac{1}{3}\big(v_2(1) + v_2(2) + v_2(1)\big) = \frac{1}{3} = \mathrm{hol}_{\mathrm{norm}}(\{1,2\}).$$

Predicted 0.333, measured 0.333; relative error 0. **Match.**

4.2 Recursion: Long-Run Contraction Rate

The iterated transfer operator $(T_3)_*$ acting on the support measure produces a sequence of supports $S_0, S_1, \ldots$ with corresponding tail-holonomy distributions over the last K steps. Paper 1 measured the standard deviations $\sigma(K = 5) = 0.395$, $\sigma(K = 10) = 0.593$, $\sigma(K = 20) = 0.180$. The non-monotone behaviour from $K = 5$ to $K = 10$ reflects a transient regime in which short trajectories

dominate small K; the genuine contraction emerges once K is comparable to typical trajectory length. Fitting $\log \sigma(K)$ linearly between $K = 10$ and $K = 20$ gives a per-step contraction rate

$$\alpha = 0.887 < 1,$$

satisfying the Banach contraction condition required for $\mathcal{R}[f] = \lim_n \mathcal{R}^{(n)}[f]$ to exist. Predicted $\alpha < 1$, measured $\alpha = 0.887$. **Match** (threshold test).

4.3 Fractality: Extensivity of $\mathrm{hol}_{\mathrm{add}}$

The multiscale structure $\mathbb{Z}_2 = \varprojlim \mathbb{Z}/2^n$ with weights $\alpha_k = 2^{-k}$ predicts that a well-converged $\mathcal{F}$-sum is dominated by its extensive component, giving Pearson correlation $r \geq 0.9$ between $\mathrm{hol}_{\mathrm{add}}$ and trajectory length. Paper 1 measured $r = 0.960$ for $n \leq 100$. Predicted $r \geq 0.9$, measured $r = 0.960$. **Match.**

The fitted slope $\mathrm{hol}_{\mathrm{add}} \approx 0.589\,|\gamma_n| + 7.523$ does not match the uniform-bit prediction $\mathbb{E}[v_2] = 0.941$ along trajectories. The discrepancy reflects that Collatz trajectories concentrate disproportionately on odd nodes (where $v_2 = 0$): the empirical slope is suppressed by the parity bias of the iteration. The match is on the qualitative correlation, not on the slope; the slope is a separate observation worth recording.

4.4 Regression: Cesàro-Mean Relaxation

The supermartingale conjecture of [9] concerns the Cesàro mean $C_k = \frac{1}{k+1} \sum_{j \leq k} \mathbb{E}_{\mu_j}[f]$. Fitting $C_k = f_\star + e^{-\lambda k}(C_0 - f_\star)$ for $N = 200$ over the full collapse trajectory:

$$C_0 = 0.667, \quad C_{\mathrm{final}} = 0.661, \quad f_\star = 0.610, \quad \lambda = 0.0011 \text{ per step.}$$

Both prediction conditions hold: $f_\star \in [0.4, 0.8]$ (the vacuum $\mathbb{E}[f]$ range, bounded by the attractor values $|1|_2 = 1$ and $|2|_2 = 0.5$) and $\lambda > 0$ (genuine relaxation). **Match.**

4.5 Synergy: Parity-Branch Coupling at the Crossing

At the Orzechowski crossing for $N = 200$, the support is $\{1, 2, 4, 5\}$ with conditional expectations $\mathbb{E}[f \mid B{=}1] = 1.000$ (odd nodes: $\{1, 5\}$) and $\mathbb{E}[f \mid B{=}0] = 0.375$ (even nodes: $\{2, 4\}$). The Synergy bilinear pairing $B(\Psi_{\mathrm{odd}}, \Psi_{\mathrm{even}})$ taken to be the absolute difference of conditional expectations evaluates to $|1.000 - 0.375| = 0.625$, which equals the Paper 1 vacuum measure M_1 at the crossing. Predicted 0.625, measured 0.625. **Match.**

We note that this test is near-tautological: the Synergy recipe and the M_1 definition produce the same arithmetic. What it verifies is that the EchoKey-prescribed coupling form lands on the right Paper 1 observable, not that it generates new information about the crossing. The test serves as a consistency check on the assignment; a non-tautological synergy test (involving a higher-order pairing that produces a quantity not directly defined by Paper 1) is identified as a follow-up in Section 6.

4.6 Refraction: Siegel Conjugacy as Layer Transform

Siegel's (p, q)-adic conjugacy [6] identifies the shortened Collatz map T_3 on $\mathbb{Z}_2$ with the 2-adic shift σ, which we model as the $L = 1$ layer transform $\Pi[\Psi; 1] = \Psi + \mu\eta(\Psi)$. The conjugacy preserves the 2-adic valuation: for any integer n, $v_2(n)$ computed by direct integer arithmetic equals $v_2(n)$ computed as the count of trailing zeros in the base-2 expansion. We tested this invariance over

the trajectories of $n = 1, \ldots, 100$, comparing the per-trajectory v_2-mean computed in the integer layer and the bit-string layer. The maximum difference across all trajectories is 0 (exactly, not just within tolerance). **Match.**

This test is a structural identity: the integer and bit-string definitions of v_2 agree by construction. What it verifies is that the EchoKey Refraction operator, applied to the Siegel conjugacy as its layer transform, preserves the invariant the conjugacy is supposed to preserve. As with the Synergy test, the result is a consistency check rather than a non-trivial computation.

4.7 Outliers: Independent Recovery of the Crossing

The Outliers operator models the Orzechowski crossing as a singular impulse in the v_2-mean trajectory. We applied a data-driven jump-detection pipeline:

1. Compute v_2-mean(k) for each step k in the $N = 200$ collapse trajectory.
2. Compute the windowed mean $\overline{v_2}_W(k)$ with $W = 5$.
3. Locate $k^* = \arg\min_k \big(\overline{v_2}_W(k+1) - \overline{v_2}_W(k)\big)$, the step with the largest negative jump.
4. At the located k^*, compute the symmetric pre/post-window means and report their difference as $\Delta\bar{v}_2$.

The pipeline is given no information about the Paper 1 crossing location. The result:

$$k^* = 76, \qquad \Delta\bar{v}_2 = -0.557.$$

Paper 1 reports the Orzechowski crossing at $k = 75$ (defined by support shrinkage to size ≤ 4) with $\Delta\bar{v}_2 = -0.480$. The location is recovered to within one step; the amplitude agrees within 14%. Predicted -0.557, measured -0.480 (tolerance 15%). **Match.**

Observation 4.1. *The Outliers test is the strongest of the seven in the sense that it is the only test that performs an independent recovery of a Paper 1 quantity not used in the prediction. The jump-location pipeline finds the crossing within one step using only the v_2-mean trajectory, without input from the support-size criterion that defines the crossing in Paper 1. This is the kind of cross-method agreement that distinguishes a structural assignment from a relabeling.*

4.8 Summary of Verification Results

Op	Predicted	Measured	Tol.	Verdict
$\mathcal{C}$	0.333	0.333	10^{-6}	MATCH
$\mathcal{R}$	$\alpha < 1$	$\alpha = 0.887$	threshold	MATCH
$\mathcal{F}$	$r \geq 0.9$	$r = 0.960$	threshold	MATCH
$\mathcal{G}$	$f_\star \in [0.4, 0.8]$, $\lambda > 0$	$f_\star = 0.610$, $\lambda = 0.0011$	threshold	MATCH
$\mathcal{S}$	0.625	0.625	10^{-6}	MATCH
Π	0	0	10^{-12}	MATCH
$\mathcal{O}$	-0.557	-0.480	15%	MATCH

Seven of seven assignments produce predictions matching their Paper 1 anchor within pre-declared tolerances. Two of the seven $(\mathcal{S}, \Pi)$ are consistency checks on the assignment recipe rather than non-trivial computations; the remaining five involve independent computations that reconstruct or match a Paper 1 anchor.

5 Realisation of the OP2 Requirements

OP2 of [9] asks for three structural objects. We identify each with a specific consequence of the seven-operator decomposition:

(i) **The basepoint in $\pi_1^{\mathrm{un}}(\mathrm{Spec}(\mathbb{Z}_2))$.** The Cyclicity assignment selects the attractor cycle $\{1, 2\}$ as the period-3 realization $[1, 2, 1]$. The basepoint is the fixed point of $\mathcal{C}_T$ acting on the iterate sequence: in operator terms, the unique state on which the period projector acts as the identity. Empirically, this state has $\mathrm{hol}_{\mathrm{norm}} = 1/3$ and the canonical signature $M_4 = 2$ of the Paper 1 vacuum.

(ii) **The target p-adic Lie group.** The Recursion and Fractality assignments together generate the multiscale structure required: $\mathcal{R}$ provides the iterative self-similarity (with verified contraction rate $\alpha = 0.887$), and $\mathcal{F}$ provides the layered scaling $\alpha_k = 2^{-k}$ realizing the $\mathbb{Z}_2 = \varprojlim \mathbb{Z}/2^n$ tower. The natural target group consistent with these is the abelian quotient of the unipotent radical of $\mathrm{GL}_n(\mathbb{Z}_2)$ generated by the layered shifts; specifying the precise non-abelian extension is part of OP6 of [9].

(iii) **The groupoid morphisms.** The Synergy and Refraction assignments together realize the morphism structure: $\mathcal{S}$ couples adjacent parity branches (verified by the M_1 recovery), and Π provides the layer transform between the integer and shift representations (verified by exact v_2-mean invariance). A discrete Collatz step $T_3 : n \mapsto T_3(n)$ is an étale-fundamental-groupoid morphism in the sense that it factors through both: $\mathcal{S}$ governs the within-layer parity transition, and Π governs the between-layer identification.

The Regression and Outliers assignments do not address OP2's three structural requirements directly. Instead they certify dynamical properties of any holonomy lift: $\mathcal{G}$ ensures supermartingale convergence (necessary for an optional-stopping argument on the lifted invariant), and $\mathcal{O}$ identifies the sole singular event in the Collatz collapse (the Orzechowski crossing) as an isolated impulse rather than a smooth transition, constraining the regularity class of the lift.

We emphasize the framing: this is a structural realisation of OP2's requirements consistent with the Paper 1 empirical constraints, not an internal proof in Kim's framework. The proof would require verifying that the seven-operator decomposition coincides with the étale-fundamental-groupoid structure on $\mathrm{Spec}(\mathbb{Z}_2) \setminus S$ for an appropriate finite ramification set S. This is identified as OP-EK1 in Section 6.

6 Open Problems

The following problems remain open. As in Paper 1, problems marked *[analytic]* require pen-and-paper proof, those marked *[structural]* require identification of a precise correspondence in an existing framework, and those marked *[computational]* require an extension of the present probe.

OP-EK1. *[structural]* **Internal proof in Kim's framework.** The seven-operator decomposition is consistent with every Paper 1 measurement, but it is not currently a theorem in Kim's arithmetic Chern–Simons framework. Prove that the EchoKey operator assignments correspond to specific structures in the étale fundamental groupoid $\pi_1^{\mathrm{un}}(\mathrm{Spec}(\mathbb{Z}_2) \setminus S, b)$ for an appropriate basepoint b and ramification set S, upgrading structural consistency to a Kim-internal theorem.

OP-EK2. *[analytic]* **Strengthen the Recursion test.** The contraction rate $\alpha = 0.887$ is fitted from a two-point estimate ($K = 10$ and $K = 20$). Compute tail-holonomy std at intermediate values $K \in \{12, 14, 16, 18\}$ and confirm geometric decay across the full range, or document the deviation.

OP-EK3. *[structural]* **Resolve the Fractality slope discrepancy.** The fitted slope of $\mathrm{hol}_{\mathrm{add}}$ versus length is 0.589, against the uniform-bit prediction $\mathbb{E}[v_2] = 0.941$. The discrepancy reflects parity bias in Collatz trajectories. Derive the corrected expected slope from the empirical odd/even visit ratio along trajectories and compare.

OP-EK4. *[computational]* **Non-tautological Synergy test.** The current Synergy test reproduces M_1 by construction. Replace the bilinear pairing with a higher-order pairing (an m-ary term in the sense of Question 7.2 of [7]) that predicts a Paper 1 anchor not directly defined by the test recipe.

OP-EK5. *[structural]* **Non-abelian Refraction.** The current Refraction test verifies an abelian invariant (v_2-mean). Determine the correct non-abelian extension required to recover $\mathrm{hol}_{\mathrm{add}}$ itself (rather than only its mean) under the Siegel layer transform.

OP-EK6. *[structural]* **Operator-ordering sensitivity.** Question 10.2 of [7] asks how sensitive the v2 composite $\mathcal{H}_{v2} = \mathcal{C} \circ \mathcal{R} \circ \mathcal{F} \circ \mathcal{G} \circ \mathcal{S} \circ \Pi \circ \mathcal{O}$ is to operator ordering. Determine which pairs commute on the Collatz test bed and which produce ordering-dependent invariants.

OP-EK7. *[structural]* **Lift to the full arithmetic setting.** Extend the seven-operator analysis from $\mathbb{Z}_2$ to general $\mathbb{Z}_p$, and from the shortened Collatz map to general dynamical systems on local fields. The seven assignments here are specific to $p = 2$; identifying which are universal and which are 2-adic accidents is a prerequisite to the SM bridge of subsequent work.

7 Summary

The seven operators of the EchoKey v2 specification, applied via the assignments of Section 3, produce predictions matching every Paper 1 empirical anchor within pre-declared tolerances. Five of the seven tests involve non-trivial computations: Cyclicity recovers the attractor $\mathrm{hol}_{\mathrm{norm}}$ via time-averaging, Recursion fits a long-run contraction rate $\alpha = 0.887 < 1$, Fractality verifies extensivity at $r = 0.960$, Regression fits a Cesàro-relaxation steady state in the predicted vacuum range, and Outliers independently recovers the Orzechowski crossing location at $k^* = 76$ against the Paper 1 measurement of step 75. Two tests (Synergy and Refraction) are consistency checks on the assignment recipe.

The seven-operator decomposition realises the three structural requirements of OP2 of [9]: Cyclicity selects the basepoint, Recursion and Fractality together generate the layered structure of the target p-adic Lie group, and Synergy and Refraction together realize the étale-groupoid morphism content of a discrete Collatz step. The Regression and Outliers operators certify dynamical properties of the lift (supermartingale convergence and the regularity class implied by the singular Orzechowski impulse) that any internal proof in Kim's framework will need to respect.

The result is structural consistency between the EchoKey and Kim frameworks on the Collatz test bed. The upgrade to a Kim-internal proof remains open and is recorded as OP-EK1 in Section 6.

The seven assignments and their verification status constitute the empirical content on which subsequent work proposing SM-Lagrangian identifications via the same seven operators will be built.

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